Reference-anchored asymmetric class expansion with query-adaptive evidence alignment

Junyi Zhang[a,*]

[a]*Shandong University, China*

Abstract

Class expansion combines a trained base classifier with names and few-shot prototypes for arriving classes. We propose reference-anchored asymmetric class expansion, which aligns heterogeneous evidence against fixed base references and preserves existing rankings as the vocabulary grows. Query normalization induces a sample-adaptive admission margin without an auxiliary router; independent calibration selects its global offset under a base-accuracy constraint. Across five ImageNet-R arrivals, the method improves average incremental accuracy over a matched HyCal transfer by 10.93 and 13.74 percentage points with CLIP L/14 and B/32. Independently retuned query normalization adds 3.60 and 5.29 points of novel capacity at 95% retention. Frozen source models reach 86.82% and 58.96% novel accuracy on ImageNet-Sketch after target calibration, meeting empirical retention targets in all ten partitions per encoder.

Keywords: Class expansion, Retention constraints, Incremental learning, Evidence fusion, Vision–language models, Long-tailed recognition

*Corresponding author.

Email address: cheerssom0876@gmail.com (Junyi Zhang)

1. Introduction

A recognition system accumulates unequal evidence as its category vocabulary grows. A trained base classifier encodes supervised discrimination, while arriving classes provide category names and a few labeled images. Reusing both sources requires a joint decision across base and novel classes without a task or group oracle. The central challenge is to align their evidence while preserving the discrimination already fitted to the base task.

Pretrained representations make lightweight expansion practical. SimpleCIL [1] and RanPAC [2] develop prototype and analytic readouts; BiMC [3] and HyCal [4] calibrate multimodal or heterogeneous recognition metrics. Our setting retains the learned base classifier alongside semantic anchors and novel visual prototypes. These branches have different score origins and scales. Standardizing against the active candidate vocabulary introduces a further dependence: appending a class can change branch scales and reverse an existing fused ranking, even when all existing parameters remain fixed.

We propose *reference-anchored asymmetric class expansion*. Fixed base reference banks give each branch a stable coordinate system throughout vocabulary growth. Their statistics adapt to the query, aligning learned, semantic and prototype evidence before fusion. This construction preserves existing scores while producing a query-dependent base–novel admission margin. A ridge identity, $\mu_L(x) = 1/|\mathcal{B}|$, makes the learned contribution explicit: centered class-specific responses compete with prototype evidence measured relative to base visual references. The resulting joint decision requires no auxiliary routing network.

The method separates query-adaptive alignment from the system's admis-

sion requirement. Once the readout is fixed, a single global offset sets the base–novel operating point. Independent calibration chooses this offset to meet a specified base-accuracy requirement over successive arrivals. The same separation supports transfer: source-fitted classifiers, fusion parameters and novel prototypes remain frozen while target calibration adjusts admission.

Our contributions are:

- An asymmetric readout that retains learned base discrimination and incorporates few-shot novel evidence in a vocabulary-invariant reference frame.
- An explicit query-adaptive admission mechanism, supported by the ridge mean identity, controlled centering/scaling comparisons and independently retuned fusion experiments.
- Retention-constrained expansion and frozen-model transfer, with finite-sample admission control and matched comparisons across three source datasets, two CLIP encoders and a natural rendering shift.

On ImageNet-R, the readout exceeds a matched HyCal transfer by 10.93 and 13.74 points in average incremental accuracy with L/14 and B/32. Query centering raises L/14 base–novel ranking AUROC from .924 to .977; independent fusion retuning confirms capacity gains of 3.60 and 5.29 points over uncentered fusion. On ImageNet-Sketch, target calibration gives 86.82% and 58.96% novel accuracy with empirical retention attainment in every evaluated partition. These results connect the alignment mechanism to sustained recognition and usable class expansion.

2. Related work

Long-tailed recognition addresses unequal class supervision through balanced fitting and losses [5, 6, 7]. CLIP supplies semantic image–text evidence [8], while prompt adaptation [9, 10], prototypes [11, 12] and semantic caches [13] adapt pre-trained representations to labeled data. Our setting combines these evidence types according to their availability at class arrival.

Class-incremental and few-shot incremental learning study retention across successive tasks [14, 15]. SimpleCIL [1] and RanPAC [2] show the value of pre-trained prototypes and analytic readouts. KBS approaches develop multi-feature similarity supplementation [16], sharpness-aware prototype calibration [17], adaptive exploration [18], representation-enhanced analytic learning [19], and multimodal prompt fusion [20].

Recent work aligns task-specific expert inference in MoTE [21], separates expanding feature spaces through orthogonal constraints [22], and learns invariant representations through counterfactual distribution intervention [23]. We study accuracy-constrained admission with asymmetric base and novel evidence. CLIP-guided discovery [24] additionally uses unlabeled novel data, whereas our arriving classes provide names and labeled support. FeNeC [25] supplies a clustered-representative readout for controlled frozen-feature comparisons.

BiMC [3] and HyCal [4] calibrate multimodal or heterogeneous recognition metrics. We align asymmetric evidence against fixed base references, so query adaptation and vocabulary growth have distinct roles. Calibrated seen/unseen recognition [26] supplies a related operating trade-off. For retention-constrained deployment, we specialize risk-controlled selection [27]

and bounded-mean betting [28] to class-macro base accuracy. A matched HyCal transfer evaluates recognition under the same encoder, support and data access.

3. Reference-anchored asymmetric class expansion

3.1. Vocabulary-invariant alignment of asymmetric evidence

Base classes $\mathcal{B}$ have labeled, potentially imbalanced fitting data. Arriving classes $\mathcal{N}$ provide a name and k labeled support images per class. Prediction selects from $\mathcal{B} \cup \mathcal{N}$ without a group oracle. A frozen encoder produces a normalized feature $v = f(x)$; normalized text anchors t_c average eight fixed prompts. Learned base scores are $q_L(x, b) = v^\top w_b + b_b$, fitted by inverse-frequency-weighted ridge regression. Novel prototypes are normalized support means p_n, giving $q_P(x, n) = v^\top p_n$; semantic scores are $q_T(x, c) = v^\top t_c$.

For each evidence source e, a fixed base reference bank $\mathcal{R}_e$ defines

$$\mu_e(x) = |\mathcal{R}_e|^{-1} \sum_{b \in \mathcal{R}_e} q_e(x, b), \tag{1}$$

$$\sigma_e^2(x) = |\mathcal{R}_e|^{-1} \sum_{b \in \mathcal{R}_e} (q_e(x, b) - \mu_e(x))^2. \tag{2}$$

We use base text anchors, learned columns and visual means as the text, learned and prototype references. Normalized evidence is

$$\widehat{q}_e(x, c) = \frac{q_e(x, c) - \mu_e(x)}{\max(\sigma_e(x), 10^{-6})}. \tag{3}$$

The statistics adapt to each query while remaining independent of the active novel vocabulary. Fusion uses the evidence available to each row:

$$s(x, b) = \widehat{q}_T(x, b) + a\widehat{q}_L(x, b), \qquad b \in \mathcal{B}, \tag{4}$$

$$s(x, n) = \widehat{q}_T(x, n) + u\widehat{q}_P(x, n) + d, \qquad n \in \mathcal{N}. \tag{5}$$

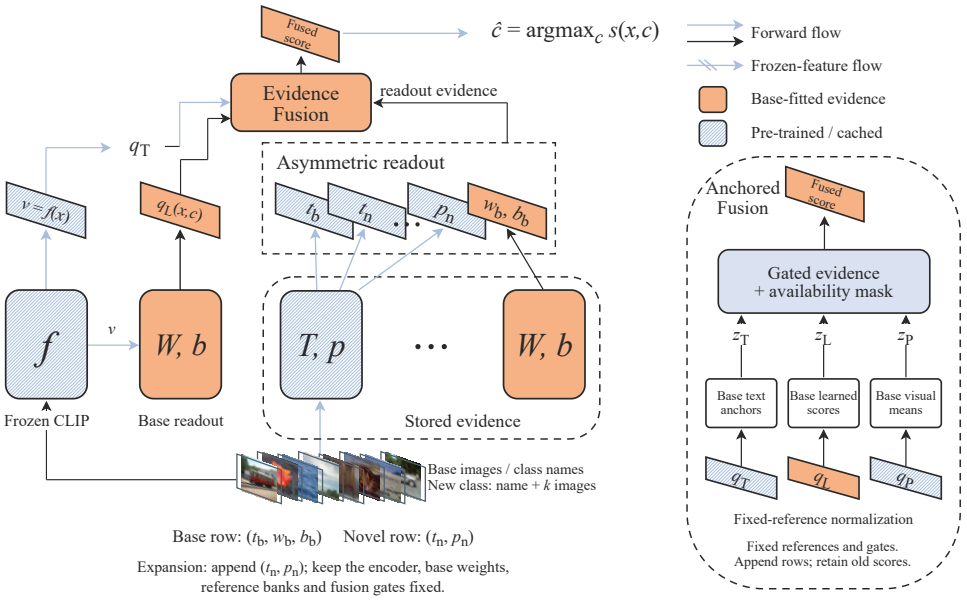


Figure 1: Reference-anchored asymmetric readout. Fixed base references set each branch's score frame; learned base columns remain stored as novel text anchors and support prototypes are appended. The separate novel offset controls admission. Orange marks base-fitted evidence; blue hatching marks pre-trained or cached evidence.

The coefficients a, u combine evidence; d controls novel admission (Fig. 1). Removing the text-centering shift shared by every candidate exposes the effective offset

$$d_{\text{eff}}(x) = d + a\frac{\mu_L(x)}{\sigma_L(x)} - u\frac{\mu_P(x)}{\sigma_P(x)}, \tag{6}$$

using the floored scales in Eq. (3). Centering adjusts the group margin for each query; scaling also changes within-group fusion.

Why the reference must be fixed. Consider old text scores $(0, 2)$ and a learned contribution $(0.25, -0.25)$. Candidate-standardized text scores $(-1, 1)$ make the second old class win after fusion. Appending a text score of 100 changes

the old standardized scores to approximately $(-0.728, -0.685)$, reversing their fused order without changing either old class's parameters. Fixed reference banks prevent this vocabulary-induced change.

Proposition 1 (Vocabulary-growth invariance)**.** *With fixed input, encoder, existing evidence, reference banks, coefficients and tie-breaking, appending classes preserves every existing score and its within-vocabulary ordering.*

Proof. Every term in Eqs. (2)–(5) for an existing class remains fixed. Hence each pairwise score difference remains fixed. □

This property concerns stored discrimination; a newly appended class can still win the expanded prediction. Raw and fixed-temperature readouts also preserve scores, providing controls that isolate query alignment.

3.2. Query-adaptive base–novel admission

Proposition 2 (Ridge identity and adaptive admission margin)**.** *Let the weighted ridge fit have one-hot targets, an unregularized intercept and a unique solution. Its learned reference mean satisfies $\mu_L(x) = 1/B$, where $B = |\mathcal{B}|$. Consequently,*

$$d_{\text{eff}}(x) = d + \frac{a}{B\sigma_L(x)} - u\frac{\mu_P(x)}{\sigma_P(x)}. \tag{7}$$

Proof. Let X include the constant feature, Y contain one-hot targets, Ω contain sample weights, and L regularize slopes. For the intercept vector e_0, $Xe_0 = \mathbf{1}$, $Le_0 = 0$ and $Y\mathbf{1} = \mathbf{1}$. The normal equations give $(X^\top \Omega X + L)W\mathbf{1} = (X^\top \Omega X + L)e_0$. Uniqueness implies $W\mathbf{1} = e_0$; summing the B learned scores gives one. Substitution into Eq. (6) completes the result. □

How alignment determines admission. The learned score contributes its deviation from the common ridge mean, $(q_L(x,b) - 1/B)/\sigma_L(x)$. The novel prototype contributes its similarity relative to base visual references, $(q_P(x,n) - \mu_P(x))/\sigma_P(x)$. With the common text-centering term removed, their competition is

$$M(x) = \max_{n\in\mathcal{N}} \left[\frac{q_T(x,n)}{\sigma_T(x)} + u\frac{q_P(x,n) - \mu_P(x)}{\sigma_P(x)} \right] - \max_{b\in\mathcal{B}} \left[\frac{q_T(x,b)}{\sigma_T(x)} + a\frac{q_L(x,b) - 1/B}{\sigma_L(x)} \right]. \tag{8}$$

The system selects the novel group when $M(x) + d > 0$ and the base group when it is negative, with the fixed class-ID rule resolving ties. Query scales make branch responses comparable; query centers adjust their relative group evidence. The scalar d then shifts this adaptive decision to the desired retention level. The mechanism uses the existing readout responses and requires no auxiliary router or task label. Its empirical signature is improved ranking of base versus novel queries together with greater correctly classified novel admission.

Each new class needs $O(D)$ storage and $O(kD)$ prototype construction after encoding; scoring K classes costs $O(KD)$. Fixed references preserve the meaning of $M(x)$ as additional classes enter the maximization.

3.3. The retention envelope

A base-only pseudo-expansion episode splits base classes into pseudo-base and pseudo-novel groups. A classifier fitted on pseudo-base classes is combined with names and reserved support for pseudo-novel classes. Held-out base queries select a, u, d by harmonic macro accuracy,

$$H = 2A_B A_N/(A_B + A_N), \tag{9}$$

with zero assigned when both accuracies vanish. The coefficients then remain fixed across actual arrivals. Actual novel query labels and test labels are excluded from this selection; grids and split seeds are in Supplementary Section K.

Let $A_{B,\mathrm{ref}}^{\mathrm{closed}}$ denote the retained readout's closed-base macro accuracy. At retention fraction $0 < r < 1$, every readout receives the same absolute base target: a 90% reference gives an 85.5% target at 95% retention. Empirical expansion capacity is

$$N_r = \max_d A_N(d) \quad \text{subject to} \quad A_B(d) \geq r A_{B,\mathrm{ref}}^{\mathrm{closed}}. \tag{10}$$

For a fixed readout, let h_B, h_N be the best labels within each group and $m(x)$ their score difference before adding d. Increasing d switches each query from h_B to h_N once, at $m(x)$, subject to the fixed tie rule. Thus $A_B(d)$ is nonincreasing and $A_N(d)$ is nondecreasing. The largest feasible event maximizes Eq. (10) for retained and replacement readouts. Test-label sweeps describe capacity; independent calibration selects deployable thresholds.

Corollary 1 (Preserved base error structure)**.** *Let R indicate correctness of the fixed retained readout restricted to base classes and C_d correctness after novel admission. Then $C_d \leq R$ on base queries. Consequently recovery $G = \Pr(R = 0, C_d = 1) = 0$, and all destruction $D = \Pr(R = 1, C_d = 0)$ goes to novel labels:*

$$A_B(d) = A_{\mathrm{ref}} - D. \tag{11}$$

The reference in this statement uses the same frozen fusion coefficients. Independently retuned readouts are compared experimentally against the common original reference, relative to which recovery is allowed.

3.4. Selecting an offset under a retention requirement

Calibration converts the aligned readout into an admission policy. Let R indicate correctness of the original closed-base reference and C_d correctness after expansion. For destruction D and recovery G, $A_B = A_{\text{ref}} - D + G$. With $\alpha = 1 - r$, define

$$L_d = R(1 - C_d) + (1 - r)(1 - R). \tag{12}$$

The condition $\mathbb{E}L_d \leq \alpha$ guarantees $A_B \geq rA_{\text{ref}}$. For the original retained readout, $G = 0$ makes these requirements equivalent. We specialize bounded-loss concentration within the Learn then Test framework [27] to unequal class-macro calibration weights.

Theorem 1 (Macro protection and tail coverage)**.** *For independent class-stratified losses in* $[0, 1]$*, let* $w_i = 1/(Bn_{c(i)})$*,* $b = \max_i w_i$ *and* $q = \sum_i w_i L_i$*. A valid p-value for* $\sum_i w_i \mathbb{E}L_i \geq \alpha$ *is*

$$p(q) = \begin{cases} \exp[-\operatorname{kl}(q, \alpha)/b], & q < \alpha, \\ 1, & q \geq \alpha. \end{cases} \tag{13}$$

Passing this bound at level δ/T *requires*

$$n_{\text{eff}} = B \min_c n_c \geq \frac{\log(\delta/T)}{\log r}. \tag{14}$$

The weighted Chernoff proof is in Supplementary Section J; the necessary coverage condition follows from the bound's minimum $p(0) = r^{1/b}$. Actual admission also depends on observed losses.

We adapt bounded-mean betting [28] to test the accuracy requirement directly through $Z_i = C_{d,i} - rR_i$.

Theorem 2 (Direct paired accuracy test)**.** *For independent correctness pairs and any fixed* $0 \le \lambda < 1/r$*,*

$$E_\lambda = \prod_i [1 + \lambda(w_i/b)Z_i] \tag{15}$$

is nonnegative with expectation at most one under $\sum_i w_i \mathbb{E}Z_i \le 0$*. A fixed convex mixture* E *gives a valid fixed-sample p-value* $\min(1, 1/E)$*.*

Supplementary Section N gives the proof and twelve fixed mixture components. This test uses the same accuracy requirement for every readout, including changes that recover reference errors.

Threshold candidates are fixed using tuning data and tested from conservative to permissive. Fixed-sequence testing stops at the first failure at level δ/T, returning the last accepted offset; if none passes, the original reference supplies zero novel admission. A union bound controls unsafe acceptance over $T = 5$ prespecified arrivals with probability at most δ. This confidence statement is per fixed method/order/partition/policy under independent target class-conditional sampling, with calibration excluded from fitting and tuning. The same calibration images serve all methods. Section P gives the conditional error-structure and variance analysis.

4. Experiments

4.1. Protocol and comparisons

We evaluate CIFAR-100 [29], Oxford-IIIT Pets [30] and ImageNet-R [31], using frozen OpenAI CLIP ViT-L/14 and ViT-B/32 features, five-shot support, ten frequency-stratified class partitions and two five-session arrival orders. Base/novel counts are 40/40, 15/15 and 100/100; each session adds eight,

three or 20 classes. Base pools follow long-tailed sampling. ImageNet-R uses task-specific splits after removing byte-identical duplicates. Novelty is relative to task fitting, with CLIP's pre-training exposure retained. Detailed pools, preprocessing and seeds are in Supplementary Sections C, I and K.

Every comparator shares its encoder, partition and support. We compare fixed-gate classifier removal, independently recalibrated removal, dense anchored text–prototype fusion, summed/class-mean Tip-Adapter caches, and raw/fixed-temperature retained fusion. The latter uses a single base-validation scale per branch. HyCal uses the author's heterogeneous cosine/Mahalanobis metrics, count weighting and Gaussian-CDF calibration, transferred to the same eight-prompt anchors and frozen encoders. Its covariance regularization and novel offset are selected by base-only pseudo-expansion. Supplementary Section M documents the pinned implementation, numerical fixtures and transfer settings. All mixtures use base-only selection; FeNeC and earlier three-partition auxiliary studies are retained in Supplementary Sections A–H.

Orders are averaged within a partition. Means and standard deviations describe ten class assignments on fixed images. Paired differences use identical cells; constrained comparisons use jointly feasible sets. Conditional paired 95% bootstrap intervals resample partition means 20,000 times. Average incremental accuracy is the mean macro accuracy over five arrivals. These intervals characterize class-assignment variation, conditional on the fixed data and encoders.

4.2. Recognition across successive class arrivals

Table 1 evaluates joint base/novel recognition across five arrivals. With L/14 and B/32, the retained readout reaches 87.29% and 67.93% average

incremental accuracy on ImageNet-R, exceeding the matched HyCal transfer by 10.93 and 13.74 points. It also exceeds this transfer on CIFAR and Pets with both encoders. Dense prototypes and fixed-temperature fusion provide close controls for the retained weights and query-dependent alignment. Table 1 uses the independently fitted source models subsequently used for admission and transfer (Supplementary Section J).

Table 1: Average incremental macro accuracy across five arrivals (%). Fusion and offsets are selected by base-only pseudo-expansion and remain fixed. Each cell gives mean ± sample standard deviation over ten partitions after averaging two orders. All readouts share frozen encoders, fitting/tuning data and five-shot novel support. HyCal is the matched transfer described in Section M. Bold marks the largest mean in each column and encoder.

Readout	CIFAR-100	Pets	ImageNet-R
CLIP ViT-L/14			
Retained readout	**83.30** ± 1.11	91.32 ± 3.44	**87.29** ± 0.56
Dense prototypes	82.40 ± 1.20	**93.09** ± 1.54	87.14 ± 0.49
Fixed temperature	81.53 ± 0.65	92.41 ± 2.37	86.31 ± 0.37
HyCal transfer	73.57 ± 2.08	83.66 ± 4.46	76.36 ± 0.58
CLIP ViT-B/32			
Retained readout	**71.73** ± 1.63	81.14 ± 3.37	**67.93** ± 0.33
Dense prototypes	71.00 ± 2.38	**81.35** ± 3.97	66.97 ± 0.69
Fixed temperature	70.22 ± 1.84	74.04 ± 11.81	66.81 ± 0.36
HyCal transfer	62.67 ± 1.32	68.26 ± 3.75	54.19 ± 0.64

Stability during vocabulary growth. The fixed-reference design isolates added-class competition from changes to existing evidence. Figure 2 shows zero measured base-score drift and zero within-base ranking churn in the three-partition vocabulary-growth study. Candidate normalization changes branch scales and can alter the existing fused ranking. Newly appended classes can still win against old classes; the preserved scores ensure that this competition uses a consistent reference frame. Five-session checks also give identical final predictions across the two arrival orders under fixed class-ID tie breaking (Section A).

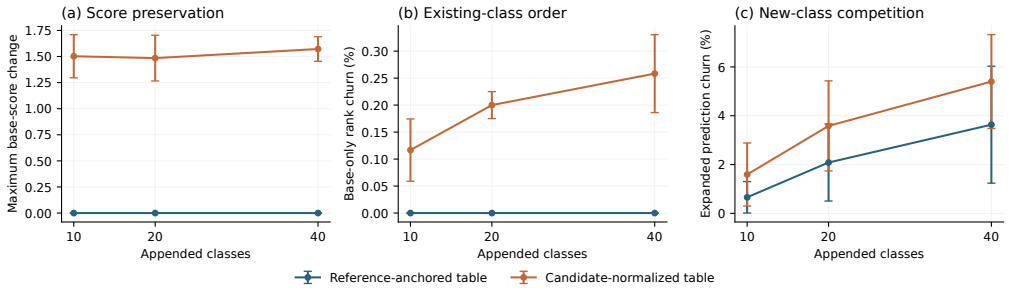


Figure 2: Score and ranking stability as the vocabulary grows, using five-shot novel prototypes. Curves average three CIFAR class partitions; bars show sample standard deviation. Fixed references preserve existing base scores and base-only rankings, while expanded predictions can change when a new class wins. This mechanism study is detailed in Supplementary Section A.

The ten-partition paired study in Section I additionally quantifies the retained classifier's incremental value: ImageNet-R gains over dense prototypes are 0.71 [0.46,0.92] and 0.90 [0.69,1.11] points for L/14 and B/32. The controlled source tables above and the full paired results retain dataset-specific comparisons, including the strong prototype performance on Pets.

Auxiliary FeNeC and analytic-readout comparisons are in Sections A–H.

4.3. Query-adaptive alignment improves base–novel decisions

The margin in Eq. (8) links reference alignment to recognition. On the original frozen ImageNet-R models, adding query centering raises base–novel ranking AUROC from .924 to .977 with L/14 and from .856 to .904 with B/32. The diagnostic uses the combined decision margin. Section P reports the separate offset distributions and all six conditions.

Table 2 isolates centering while holding query scales and fusion coefficients fixed and independently reselecting the global offset with the same direct accuracy test. On CIFAR L/14, novel accuracy rises from 30.27% to 72.00%, a paired gain of 41.73 [37.85,45.34] points. ImageNet-R L/14 rises from 62.20% to 81.61%, a 19.41 [15.09,23.45]-point gain. All six conditions show positive paired gains and every final test-retention target is met. The table reports finite admission so that fallback remains visible.

Independent fusion optimization. We also optimize the fusion coefficients and tuning offset separately for each normalization scheme under equal search budgets. At the common 95% base target, query normalization exceeds uncentered fusion by 3.60 [2.55,4.74] points on ImageNet-R L/14 and 5.29 [3.74,7.25] on B/32. Gains over fixed global normalization are 2.84 [2.42,3.28] and 4.65 [3.57,5.71] points. These capacity comparisons average both orders within ten jointly feasible partitions. Section Q reports the full readout-by-normalization comparison; Section M separately isolates centering and scaling. Together with Table 2, they establish the alignment effect both within a fixed readout and after competing fusions are independently selected.

Table 2: Query-centering ablation with fixed fusion coefficients and query scales. Query/None denotes centered/uncentered evidence; each receives independent offset calibration with identical cap-40 images and the direct test. Base/Novel are final-session test percentages, including zero novel accuracy on reference fallback. Finite counts admission out of ten partitions; all empirical retention targets are met. Paired gains (pp) have conditional 95% intervals. First arrival order.

Data	Centering	Finite	Base (%)	Novel (%)	Gain (pp)
CLIP ViT-L/14					
CIFAR-100	Query	10/10	86.01	72.00	+41.73 [37.85,45.34]
	None	10/10	85.86	30.27	
Pets	Query	10/10	94.40	87.52	+17.88 [5.58,34.99]
	None	9/10	94.39	69.65	
ImageNet-R	Query	10/10	88.63	81.61	+19.41 [15.09,23.45]
	None	10/10	88.44	62.20	
CLIP ViT-B/32					
CIFAR-100	Query	10/10	76.74	52.54	+27.10 [18.60,34.32]
	None	10/10	77.12	25.44	
Pets	Query	9/10	88.59	54.58	+25.52 [11.17,38.52]
	None	10/10	88.44	29.07	
ImageNet-R	Query	10/10	72.42	52.06	+12.23 [9.01,16.09]
	None	10/10	72.44	39.83	

Recognition at shared retention requirements. Figure 3 evaluates the ten-partition replication across five base-retention levels. At 95% retention, the ImageNet-R L/14 retained readout reaches 84.31% novel accuracy at 86.42% base accuracy. Test-label sweeps characterize the recognition available at each requirement; the next section selects offsets using independent calibration.

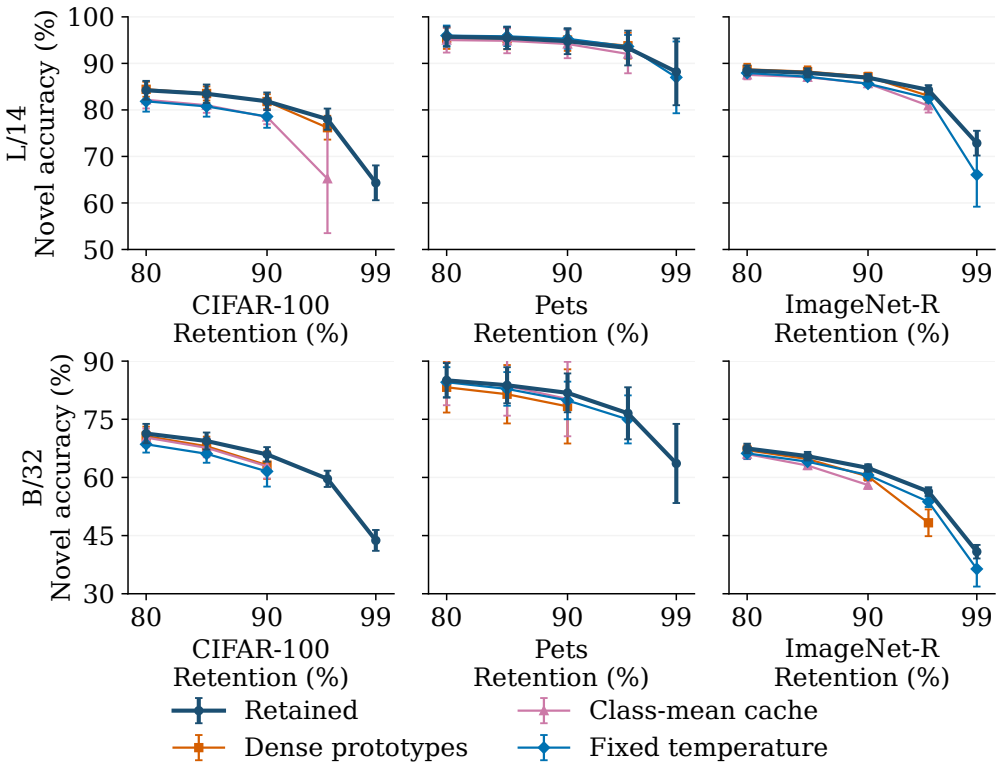


Figure 3: Novel recognition at shared base-retention requirements in the ten-partition replication (Supplementary Section I), after the fifth arrival. Each readout attains the stated fraction of the retained reference's closed-base accuracy. Means and sample standard deviations average two orders within each partition; points require all ten partitions to be feasible. Vertical ranges are 50–100% (L/14) and 30–90% (B/32); bars stop at panel limits. Test-label sweeps describe capacity. Section I gives partial feasible sets and counts.

4.4. Useful admission under a base-accuracy requirement

Independent calibration selects the global offset after fitting and base-only tuning. All readouts share calibration images, $r = .95$, $\delta = .05$ and five prespecified arrivals. Table 3 applies the direct accuracy test to every method. On ImageNet-R L/14, the retained readout reaches 81.61% novel accuracy with 10/10 admissions, compared with 76.76% for dense prototypes and 78.16% for fixed temperature. The paired retained-minus-dense gain is 4.85 [2.77,6.93] points. HyCal's matched transfer receives no certified novel admission in this condition.

On CIFAR L/14, the retained readout reaches 72.00% novel accuracy with 10/10 admissions; dense prototypes reach 36.60% with 6/10. This 35.40-point policy gain includes the ability to certify expansion. Capacity and selected-policy performance are linked through the operating trajectories in Fig. 4. For ImageNet-R L/14, the common-floor capacity difference is 0.29 [−0.55,1.16] points; matching both frontiers at the dense policy's stricter realized base accuracy gives 3.10 [1.37,4.88] points. Section P reports the complete policy/capacity decomposition, feasibility and exact discrete matching.

Calibration coverage and retained error structure. For the original retained readout, $G = 0$ makes protection coincide with the retention requirement. At cap 40, this certificate admits the final session in all ten partitions of every dataset/encoder, with all empirical test targets met. L/14 novel accuracy is 72.76% on CIFAR, 88.51% on Pets and 80.74% on ImageNet-R. All 20 arrival paths meet every test target; finite admission holds throughout CIFAR and ImageNet-R and 17/20 L/14 and 16/20 B/32 Pets paths, with early fallback

Table 3: Novel accuracy under a common direct accuracy certificate, fifth arrival, first order. Each cell gives the test mean (%) over all ten partitions and the finite-admission count in parentheses. Reference fallback contributes zero novel accuracy. All methods use identical cap-40 calibration images, 95% retention and 95% confidence over the five prespecified arrivals within a fixed configuration. All empirical test targets are met except one dense B/32 ImageNet-R case. Absolute base accuracy and admission-conditioned means are in Section N. Bold marks the largest mean.

Readout	CIFAR-100	Pets	ImageNet-R
CLIP ViT-L/14			
Retained readout	**72.00** (10/10)	**87.52** (10/10)	**81.61** (10/10)
Dense prototypes	36.60 (6/10)	71.64 (8/10)	76.76 (10/10)
Fixed temperature	59.04 (9/10)	77.43 (9/10)	78.16 (10/10)
HyCal transfer	0.00 (0/10)	3.49 (1/10)	0.00 (0/10)
CLIP ViT-B/32			
Retained readout	**52.54** (10/10)	**54.58** (9/10)	**52.06** (10/10)
Dense prototypes	23.77 (6/10)	13.25 (2/10)	23.88 (7/10)
Fixed temperature	46.52 (10/10)	44.66 (9/10)	46.35 (10/10)
HyCal transfer	0.00 (0/10)	0.00 (0/10)	0.00 (0/10)

in the remaining Pets paths. Section N compares both certificates under identical access, including dense L/14 ImageNet-R's 48.58% protection result and 76.76% direct result.

Equation (14) gives the necessary effective count 89.78 for the protection bound. Nested per-class caps 2, 5, 10, 20 and 40 show how coverage supports

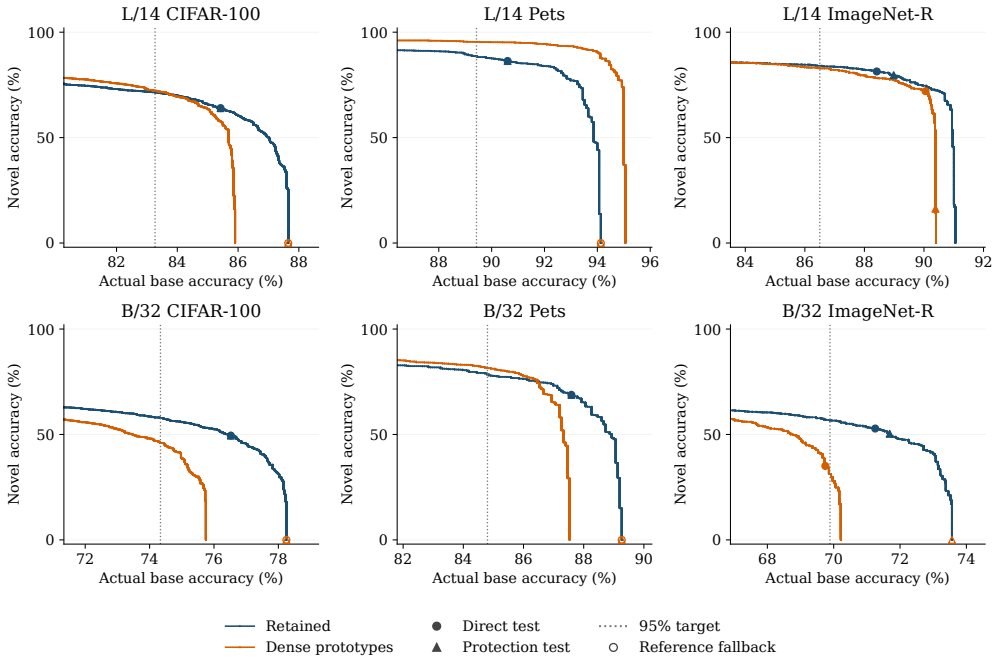


Figure 4: Recognition trajectories and independently calibrated operating points, prespecified partition 0 after the fifth arrival. The vertical line is the shared minimum base target. Triangles use protection calibration, circles direct calibration, and hollow markers denote reference fallback. Test labels describe the curves; independent calibration selects the points. Matching uses exact discrete events. Full results are in Section P.

admission (Section L). Cap 40 uses 1,600 base labels on CIFAR, 600 on Pets and 3,636–3,778 on ImageNet-R, additional to five-shot novel support. All readouts and tuning parameters remain fixed across these label budgets.

4.5. Frozen-model transfer through target calibration

The transfer study evaluates the original ImageNet-R readouts on matching ImageNet-Sketch classes [32]. The encoder, base classifier, fusion coefficients, five-shot source prototypes and candidate offsets remain fixed. Source and

target calibration receive equal per-class counts; target labels select the scalar admission offset. Identity-disjoint target pools are fixed before scoring (Supplementary Section R).

With direct target calibration, retained readouts attain 86.82% novel accuracy at 90.76% base accuracy with L/14 and 58.96% at 78.87% with B/32. All five sessions admit expansion and meet their empirical test-retention targets in every partition. Dense L/14 reaches 83.34% novel accuracy with 10/10 final admissions; dense B/32 reaches 8.35% including fallback, with 3/10 admissions and 27.84% accuracy conditional on admission. These results demonstrate useful transfer of the source readout through target-domain admission calibration.

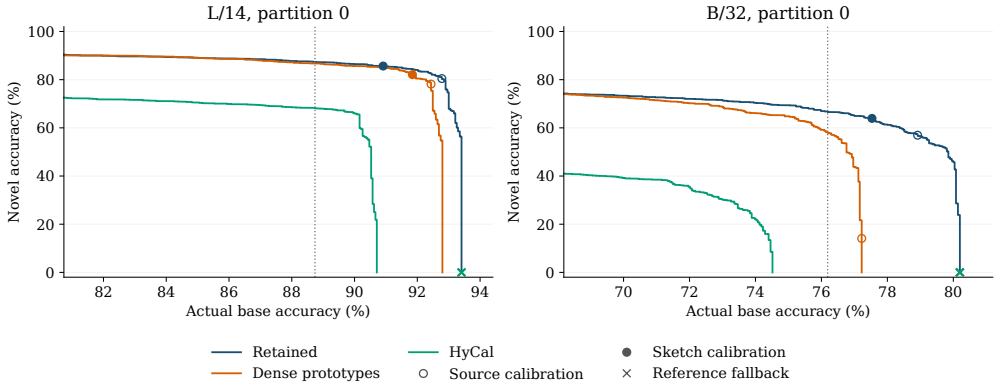


Figure 5: Frozen-source recognition on ImageNet-Sketch, partition 0. Open/filled circles use source/target calibration with equal per-class counts; crosses denote reference fallback. Dotted lines show the base requirement. Test-label trajectories describe capacity; calibration selects the operating point. All ten partitions, absolute accuracies and admission counts are in Section R.

Source calibration also meets all retained test targets, at 82.99% and 54.87% novel accuracy. Target calibration therefore increases admitted novel

recognition in this natural-shift study. The separate CIFAR noise/blur stress test shows the need for calibration–query correspondence: clean calibration meets 5/10 and 1/10 L/14 targets, whereas matched-shift calibration restores 10/10 with finite admission for both encoders (Section O). Population confidence applies to the target class-conditional distribution represented by the calibration data.

5. Conclusion

Reference-anchored asymmetric class expansion preserves learned base discrimination while incorporating semantic and few-shot novel evidence. Fixed base references eliminate vocabulary-induced score changes; query normalization aligns heterogeneous responses and creates an explicit sample-adaptive admission margin. Controlled experiments connect this mechanism to stronger base–novel decisions and sustained incremental recognition. A separately calibrated scalar offset enforces a base-accuracy requirement and transfers frozen source models to a natural rendering shift. The method provides a coherent path from asymmetric evidence to stable, useful class expansion.

CRediT authorship contribution statement

Junyi Zhang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review and editing.

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Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The experiments use public CIFAR-100, Oxford-IIIT Pets, ImageNet-R and ImageNet-Sketch data and the publicly released CLIP ViT-L/14 and ViT-B/32 checkpoints. Code, source-hashed experimental records, prediction archives, manuscript sources, and table/figure generation scripts are available at `https://github.com/Cheerssom/Reference-Anchored-Candidate-Tables`. Dataset images, pretrained weights, and feature caches are obtained separately using the repository instructions.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

OpenAI Codex was used to assist with literature discovery, manuscript drafting and revision, experiment and verification scripts, and LaTeX and table preparation. The author is responsible for the accuracy, interpretation and originality of the submitted work and for reviewing and approving the final manuscript and supporting materials.

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